

Trainings ▾

General Information

Introduction

This procedure describes the proper method for placing an engine into long-term storage (more than 6 months), and returning a stored engine to service. The storage procedure applies to engines in running condition and applies whether the engine will remain in the equipment or will be removed for storage.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

CAUTION

Never use diesel fuel containing bio components for internal preservation of fuel system for engine storage. Fuel degradation may cause damage and lead to malfunction of fuel system components.

Preparing Spare or Rebuilt Engine for Long Term Storage

The steps below **only** apply to engines that are received directly from a Cummins Inc. assembly plant or a Master Rebuild Center.

- Apply a lightweight lubricant or rust preventative to machined surfaces that are **not** painted.
- Apply a lightweight lubricant or rust preventative to any internal cast surfaces, such as coolant inlet and outlet connections, that are visible and **not** painted.
- Cover engine and component openings with clean care caps or heavy paper and tape to prevent dirt and moisture from entering engine.
- Remove all external drive belts.
- Install appropriate desiccant on engine to absorb any moisture.
- Cover entire engine with plastic.
- Move engine or equipment to an appropriate storage area.
 - Storage area **must** be dry with uniform temperature. Recommended storage temperature is between -12°C [10°F] and 49°C [120°F], with relative humidity less than 60 percent.

Preparing an In-Use Engine for Long Term Storage

To prepare an engine for long term storage, complete the following steps. Reference corresponding service procedures as necessary.

- If engine will be stored in equipment:
 - Change engine lubricating oil.
 - Replace engine lubricating oil filters.
 - Verify cooling system is filled with an approved, fully formulated coolant.
 - Measure supplemental coolant additive (SCA) concentration using Fleetguard® coolant test kit CC2602 and adjust, if necessary.
 - Check coolant quality annually during storage.
 - Replace fuel filters.
 - Verify fuel supply in the fuel tank contains pure diesel fuel. (0 percent biofuel content).
 - If there is any biofuel present in the fuel system, the biofuel will degrade the system while in storage.
 - Start engine and bring engine coolant to operating temperature. Run engine for 30 minutes.
 - For further information on engine operating instructions, see corresponding Operation and Maintenance Manual. Reference Section 1.
 - Turn engine off.
- If engine is to be removed from equipment, but is still presently installed:
 - Verify fuel supply in the fuel tank contains pure diesel fuel (0 percent biofuel content).
 - If there is any biofuel present in the fuel system, the biofuel will degrade the system while in storage.
 - Start engine and bring engine coolant operating temperature. Run engine for 30 minutes.
- If engine has been removed from equipment:
 - Drain engine lubricating oil.
 - Drain coolant.
 - Apply a lightweight lubricant or rust preventative to machined surfaces that are **not** painted.

- Apply a lightweight lubricant or rust preventative to any internal cast surfaces, such as coolant inlet and outlet connections, that are visible and **not** painted.
- If engine is equipped with a marine heat exchanger:
 - Drain raw water system
 - Flush raw water system
 - Drain water
- Cover all of engine and component openings with clean care caps or heavy paper and tape to prevent dirt and moisture from entering engine.
- Remove all external drive belts.
- Cover entire engine with plastic.
- Move engine or equipment to an appropriate storage area.
 - Storage area **must** be dry with uniform temperature. Recommended storage temperature is between -12°C [10°F] and 49°C [120°F], with relative humidity less than 60 percent.

Preparing Aftertreatment Components for Long Term Storage

- For Engine with a displacement of 19 L and above refer to Service Bulletin, Storage of Individual UL2 Aftertreatment Diesel Exhaust Fluid (DEF) System Components, Bulletin 5504262 (/qs3/pubsys2/xml/en/bulletin/5504262.html) for instructions on preparing aftertreatment components for long term storage. Refer to procedure 011-143 in Section 11 for instructions on Purging the Diesel Exhaust Fluid in preparation for storage.

Refer to procedure 011-143 Aftertreatment Diesel Exhaust Fluid Dosing Unit Manual Purge, for instructions on Purging the Diesel Exhaust Fluid in preparation for storage

Removing Engine from Long Term Storage

To remove the engine from long term storage, complete the following steps below. Also reference corresponding Service Manual procedures.

- Install new fuel and coolant filters, if equipped.
- Fill engine to high mark with engine lubricating oil.
- Cooling system procedures:
 - If engine cooling system was drained:
 - Fill cooling system with coolant. See Coolant Recommendations and Specifications procedure in Section V of corresponding Owners or Operation and Maintenance manual for cooling system specifications.
 - If engine was stored with fully formulated coolant:
 - Check coolant quality. Replace coolant if necessary.
 - Measure SCA concentration and adjust, if necessary.
 - Verify cooling system is properly filled. See Coolant Recommendations and Specifications procedure in Section V of corresponding Owners or Operation and Maintenance manual for cooling system specifications.
- Prime lubricating oil system.
 - With Pre-lubricating Oil Pump:
 - Complete a pre-lubricating oil system cycle.
 - Rotate crankshaft 240 degrees.
 - Complete a pre-lubricating oil system cycle.

- Rotate crankshaft 300 degrees.
- Complete a pre-lubricating oil system cycle.
- Rotate crankshaft 240 degrees.
- Complete a pre-lubricating oil system cycle.
- Without Pre-lubricating Oil Pump:
 - Install oil pressure gauge or pressure transducer in the engine lubricating oil system as far downstream of the lubricating oil pump as possible.
 - Crank engine without firing for 30 seconds while observing oil pressure.
 - Pause for 3 minutes.
 - Repeat steps 2 and 3 above until nominal oil pressure is observed.
- Install all external drive belts.
- Replace all spark plugs, if applicable.
- If engine is equipped with a marine heat exchanger:
 - Open raw water supply valves
 - Prime raw water pump
- Verify all fuel supply hoses have been tightened.
- Verify all fuel supply and return shutoff valves are open.
- Install all aftertreatment components, if applicable.
- Start engine. See Normal Starting Procedure in Section 1 of corresponding Owners or Operation and Maintenance Manual.
- For electronically controlled engines, troubleshoot any active fault codes.

Removing Aftertreatment Components for Long Term Storage

For Engine with a displacement of 19 L and above refer to Service Bulletin, Storage of Individual UL2 Aftertreatment Diesel Exhaust Fluid (DEF) System Components, Bulletin 5504262 (</qs3/pubsys2/xml/en/bulletin/5504262.html>) for instructions on preparing aftertreatment components for removing from long term storage.